Applied Statistics Assignment

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1 Linear Models

Assuming the NLM with $y = X\beta + \epsilon$ and $\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, we have the likelihood function

$$L(\beta, \sigma^2) := \frac{1}{(2\pi\sigma^2)^{\frac{n}{2}}} \exp\left(-\frac{(y - X\beta)^T(y - X\beta)}{2\sigma^2}\right) \tag{1}$$

so that maximising (1) is equivalent to minimising twice the negative log-likelihood given by

$$l(\beta, \sigma^2) := n\log(2\pi) + n\log(\sigma^2) + \frac{(y - X\beta)(y - X\beta)}{\sigma^2}. \tag{2}$$

1.1 Properties of the MLE

(a) $\hat{\beta} := (X^TX)^{-1}X^Ty$ is an unbiased MLE for β since

$$\begin{aligned}
\mathbb{E}[\hat{\beta}] &= \mathbb{E}[(X^TX)^{-1}X^Ty] \\
&= \mathbb{E}[(X^TX)^{-1}X^T(X\beta + \epsilon)] \\
&= (X^TX)^{-1}X^TX\beta + (X^TX)^{-1}X^T\mathbb{E}[\epsilon] \\
&= \beta.
\end{aligned}$$

(b) The covariance matrix for $\hat{\beta}$ is given by

$$\begin{aligned}
Cov(\hat{\beta}) &= \mathbb{E}[(\hat{\beta} - \mathbb{E}[\hat{\beta}])(\hat{\beta} - \mathbb{E}[\hat{\beta}])^T] \\
&= \mathbb{E}[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] \\
&= \mathbb{E}[\hat{\beta}\hat{\beta}^T] - \beta\mathbb{E}[\hat{\beta}]^T - \mathbb{E}[\hat{\beta}]\beta^T + \beta\beta^T \\
&= \mathbb{E}[\hat{\beta}\hat{\beta}^T] - \beta\beta^T \\
&= \mathbb{E}[((X^TX)^{-1}X^Ty)((X^TX)^{-1}X^Ty)^T] - \beta\beta^T \\
&= \mathbb{E}[((X^TX)^{-1}X^T\epsilon)((X^TX)^{-1}X^T\epsilon)^T] \\
&= \sigma^2((X^TX)^{-1}X^T)((X^TX)^{-1}X^T)^T \\
&= \sigma^2(X^TX)^{-1}X^TX(X^TX)^{-1} \\
&= \sigma^2(X^TX)^{-1}
\end{aligned}$$

(c) Since $\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ and $\hat{\beta} = (X^T X)^{-1} X^T (X\beta + \epsilon)$ is a linear function of ϵ, $\hat{\beta}$ must be normally distributed with mean and covariance as computed in parts a) and b) respectively.

1.2 MLE for Variance in NLMs

In order to compute the MLE for σ^2, we differentiate (2) to find the first order optimality condition

$$\left.\frac{\partial l}{\partial(\sigma^2)}\right|_{\hat{\sigma}^2, \hat{\beta}} = \frac{n}{\hat{\sigma}^2} - \frac{(y - X\hat{\beta})^T (y - X\hat{\beta})}{(\hat{\sigma}^2)^2} = 0$$

which gives

$$\hat{\sigma}^2 = \frac{(y - X\hat{\beta})^T (y - X\hat{\beta})}{n}.$$

This is the mean squared error of our NLM.

1.3 Exponential Family of Distributions

The exponential family of distributions consists of all distributions with pdfs of the form

$$f(x; \theta, \phi) = \exp\left(\frac{x\theta - b(\theta)}{a(\phi)} + c(x, \phi)\right) \tag{3}$$

where a, b, c are functions to be chosen and ϕ is the dispersion parameter.

(a) For the Poisson distribution,

$$\begin{aligned} f(x) &= \frac{\mu^x e^{-\mu}}{x!} \\ &= \exp\left(x \log(\mu) - \mu - \log(x!)\right) \end{aligned}$$

so that the Poisson distribution is exponential with

$$\begin{aligned} &\theta = \log(\mu), \\ &\phi = 1, \\ &a(\phi) = \phi, \\ &b(\theta) = e^{\theta}, \\ &c(x, \phi) = -\log(x!) \end{aligned}$$

as in (3).

(b) For the exponential distribution,

$$\begin{aligned} f(x) &= \mu e^{-\mu x} \\ &= \exp(\log(\mu) - \mu x) \end{aligned}$$

Cluster	Setosa	Versicolor	Virginica
1	0	39	14
2	50	0	0
3	0	11	36

Figure 1: Contingency table for k means clustering on the iris dataset.

so that the exponential distribution is a member of the exponential family of distributions with

$$\begin{aligned}\theta &= \mu,\\ \phi &= 1,\\ a(\phi) &= -\phi,\\ b(\theta) &= \log(\theta),\\ c(x,\phi) &= 0\end{aligned}$$

as in (3).

2 Clustering

2.1 Number of Clusters

We choose to separate our data into 3 clusters since there are 3 classes of plant to classify. Although the data is more clearly separable into 2 main clusters as $k = 2$ gives the highest silhouette score, we choose to use $k = 3$ in order to attempt to cluster by species. This results in a higher adjusted Rand index than $k = 2$ - indicating that $k = 3$ clusters more closely corresponds to the species labelling.

2.2 Comparison of Clusters with Target Classes

In order to compare our obtained clusters with the true species classification, we compute the contingency table figure 1.

This demonstrates that our clustering was able to perfectly separate species into Setosa and not Setosa, corresponding to the higher silhouette score for $k = 2$ although we are not able to separate the other two species without producing less distinct clusters. We see an adjusted Rand index of 0.568 for $k = 2$ compared to 0.620 for $k = 3$ indicating that $k = 3$ clusters more accurately match the true species classification.

2.3 Principal Component Analysis

When plotting the data against the first 2 principal components, we observe the clear separation of group 2 from groups 1 and 3. This corresponds to the correct

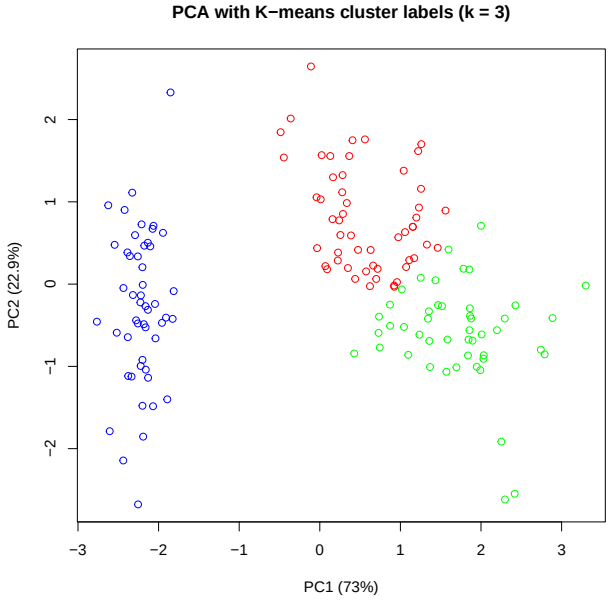


Figure 2: Principal components of the iris dataset coloured by k-means clusters with $k = 3$. Group 1: Red, Group 2: Blue, Group 3: Green.

labelling of the distinct setosa species. However, there is some overlap between groups 1 and 3, indicating some overlap in the characteristics of the versicolor and virginica species and less distinct separation between our clusters.

2.4 Derived Variables

We use the following derived variables:

- Sepal area = sepal length * sepal width: Better specifies the overall size of the sepal.
- Petal area = petal length * petal width: Better specifies the size of petals - easier to compare with sepals.
- Sepal ratio = sepal length / sepal width: Specifies the shape of the sepals
- Petal ratio = petal length / petal width: Specifies the petal shape in an easily comparable way to the sepal shape.
- Petal to sepal length = petal length / sepal length: Allows for easy comparison between petal and sepal length.
- Petal to sepal width = petal width / sepal width: Allows for easy comparison between petal and sepal width.

Cluster	Setosa	Versicolor	Virginica
1	50	0	0
2	0	9	34
3	0	41	16

Figure 3: Contingency table for k means clustering with derived variables on the iris dataset.

These choices allow us to directly compare the shapes of parts of the plant relative to each other while retaining area variables for the overall size of the plant.

In order to assess the quality of our derived variable selection, we again separate into $k = 3$ clusters despite $k = 2$ having the highest silhouette score for the derived variables. This gives the contingency table in figure 3in figure 3.

This contingency table demonstrates that again we were able to completely separate setosa from the the other species using our derived variables, although there remains some overlap between the other 2 species. The adjusted Rand index was 0.621, indicating a slightly better match with true species labels than was achieved with the original variables.

2.5 t-SNE

In figure 4, we observe that a lower perplexity of 5 creates small groupings within clusters. It is difficult to interpret which of these smaller groupings should be closer to each other to accurately reflect the cluster groupings. This is possibly due to the tendency of small perplexity values to focus on local relationships between data without considering global relationships between groups of data. When perplexity is increased to 15, these smaller groupings appear to become closer to each other and form 2 distinct clusters, although the cluster containing 2 species still fails to separate them in a visually meaningful way. At perplexity 30, we observe the best clustering performance compared to k-means, with a distinct group for cluster 1 and 2 slightly overlapping groups for clusters 2 and 3 with some level of overlap. This most accurately reflects our results from k-means clustering. When increasing perplexity to 45, our data forms 2 very distinct groups with minimal within group separation of the 2 overlapping species. This demonstrates the tendency of high perplexity values to focus on global data separation while failing to account for local relationships between data that are similar, and matches k-means clustering results with $k = 2$.

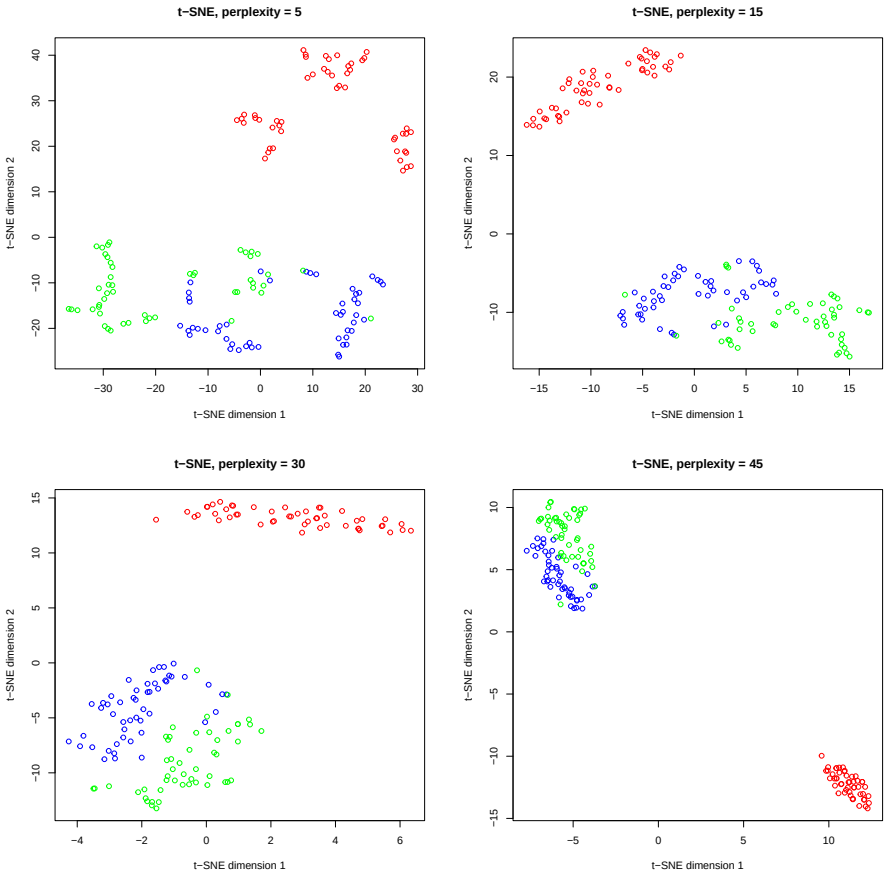


Figure 4: t-SNE visualisations of the iris dataset coloured by k-means clustering with $k = 3$ across a range of perplexity values.

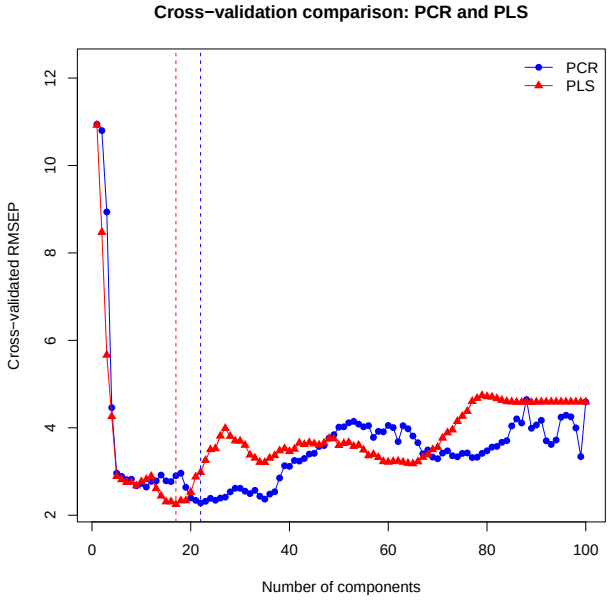


Figure 5: Predictive error during crossvalidation for PCR and PLS

3 Regression with Derived Variables

3.1 Experiment Setup

We begin by dividing our data into a 70/30 train/test split before training PCR and PLS regression on the training data using 10-fold cross validation. We then select the number of principle components corresponding to the lowest root mean square predictive error after cross validation. For PCR, we select 22 components while for PLS we select 17. The predictive RMSE is demonstrated in figure 5.

3.2 Predictive Performance

In order to compare the predictive performances of PLS with PCR, we apply our trained models to the test dataset to compute predictions. We then compare these predictions to the true target values using root mean square error (RMSE), mean absolute error (MAE) and by computing the R^2 statistic. These values are displayed in figure 6.

Overall, we observe better performance from PCR than PLS with lower RMSE and MAE values and a higher R^2 statistic. For further analysis, we

Method	Components	RMSE	MAE	R^2
PCR	22	3.385	2.049	0.937
PLS	17	4.023	2.254	0.911

Figure 6: Performance metrics for PCR and PLS regression

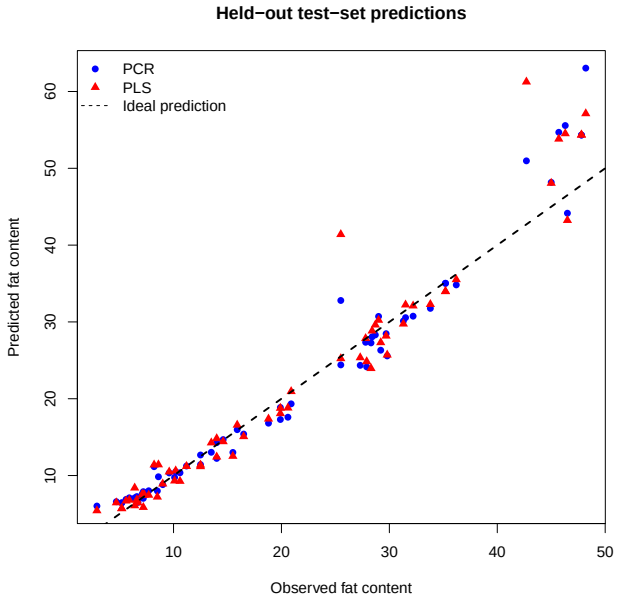


Figure 7: Observed vs predicted targets using PCR and PLS regression.

plot the predictions from each method in figure 7. it is difficult to identify any meaningful visual difference between the methods from this plot.

In order to more closely analyse the predictions plotted in figure 7, we plot the residuals of our predictive methods in figure 8. Again, we observe very little visual difference suggesting that PCR and PLS perform similarly on the data for predictive tasks, with a small advantage for PCR visible only when computing precise error metrics.

3.3 Conclusion

We conclude that PCR performs slightly better than PLS regression for predictive tasks when trained on the meatspec dataset, with PCR yielding slightly better performance on all directly computed error metrics. However, PLS was able to achieve visually similar results with a smaller number of regressors which

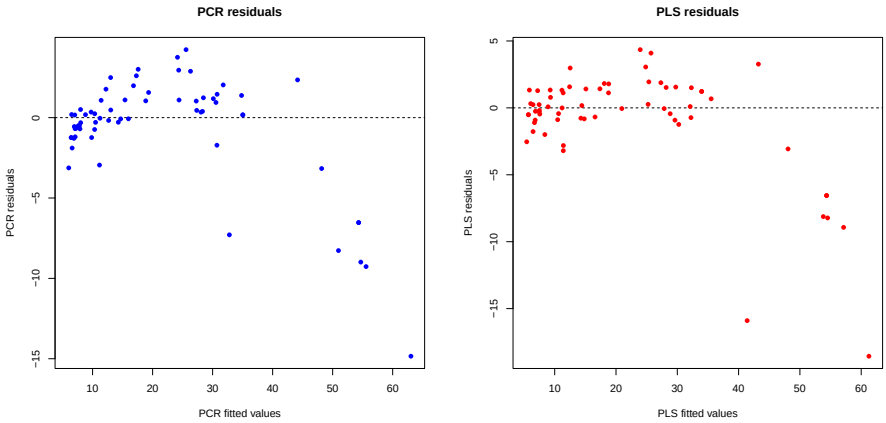


Figure 8: Residuals of predicted vs true target data using PCR and PLS regression.

suggests that it may generalise just as well to other datasets.

4 Lasso Regression

We aim to examine the effect of correlation within the design matrix on the performance of lasso regression for identifying associated variables.

4.1 Study Design

We describe the design of a study for investigating the power of lasso for identifying associated variables.

(a) In order to generate a data matrix with 200 samples and 400 variables, we create a 200×400 matrix of zeros and divide this into 40 200×10 blocks. Within each block, we generate normally distributed data with correlation strength ρ so that samples are correlated within blocks, and blocks are independent from each other. We then normalise the full data matrix to have unit variance. This gives us a suitable data matrix for investigating the effects of correlation within the design matrix.

(b) In order to generate a continuous response vector, we assume a normal linear model with

$$y = X\beta + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I_{200}) \tag{4}$$

where $\beta = (2, 1.75, 1.5, 1.25, 1, -2, -1.75, -1.5, -1.25, -1, 0, 0, ...)$ is known. This allows us to generate the response vector y by drawing suitable samples of ϵ. Since only the first 10 entries in β are non zeros, only the first 10 variables from each sample in our data are truly associated.

(c) As introduced in [4], lasso aims to solve the minimisation problem

$$\min_{\beta \in \mathbb{R}^p} \{||y - X\beta||_2^2 + \lambda||\beta||_1\}$$

with $\lambda \geq 0$ a tuning parameter. The l^1 penalty term for the coefficient β promotes sparsity in order to avoid overfitting and account for associated variables in the data. λ is known as a regularisation parameter since increasing λ leads to greater bias with lower variance, while taking $\lambda = 0$ retrieves a standard OLS regression with low bias and high variance (overfitting) [2].

In order to select λ, we use 10-fold cross-validation with MSE loss to compute the λ value for which the generalisation error is one standard error away from the minimum - the most parsimonious choice of λ as well as the value of λ which minimises MSE on the held out set. The values of β which solve the optimisation problem for these λs give us our regression coefficients.

(d) In order to evaluate the performance of the lasso regression for identifying associated variables, we repeat lasso regression on generated data in order to compute a range of performance metrics for both choices of λ across a range of correlation strengths. Defining tp, fp, tn, fn to mean the number of true positives, false positives, true negatives, false negatives respectively achieved with our classification. Repeating the experiment $N = 100$ times for each set of parameters allows us to compute the confusion matrix for each set of parameters. From here, we can compute the performance metrics

- Sensitivity: $\frac{tp}{tp+fn}$
- Specificity: $\frac{tn}{tn+fp}$
- Precision: $\frac{tp}{tp+fp}$
- Accuracy: $\frac{tp+tn}{p+n}$

We can then plot these values, along with the number of selected variables as the correlation strength varies, which will allow us to draw conclusions about the performance of lasso for this task.

4.2 Results and Analysis

We run the study described in question 4.1, with the within-block correlation strengths $\rho = 0, 0.25, 0.5, 0.75, 0.9$ while only the first 10 variables are truly associated in the response vector through the relationship (4). We compare the most parsimonious choice of λ (P) with the λ which minimises cross validation error (M) by computing the following performance metrics, averaged over 100

ρ	λ	Sensitivity	Specificity	Precision	Accuracy	Selected Variables
0.00	P	1.000	0.970	0.506	0.971	21.75
0.00	M	1.000	0.893	0.219	0.896	51.80
0.25	P	1.000	0.957	0.405	0.958	26.94
0.25	M	1.000	0.883	0.193	0.886	55.66
0.50	P	1.000	0.936	0.308	0.938	34.80
0.50	M	1.000	0.858	0.164	0.862	65.23
0.75	P	0.998	0.905	0.221	0.907	47.05
0.75	M	1.000	0.842	0.147	0.846	71.49
0.90	P	0.969	0.887	0.184	0.889	53.88
0.90	M	0.995	0.819	0.130	0.823	80.55

Figure 9: Performance metrics for lasso regression averaged over 100 initialisations of the experiment.

runs of the experiment.

These performance metrics in figure 9, visualised in figure 10 demonstrate that choosing the most parsimonious λ improves the performance of our model on held out data and leads to sparser regression coefficients for all correlation strengths. In particular, we observe greater specificity, precision and accuracy for the parsimonious choice of λ at all correlation strengths. All performance metrics decrease as ρ increases, and the regression coefficient β becomes less sparse. This indicates that lasso is less able to correctly identify associated variables with high levels of correlation within the design matrix.

4.3 Further Effects on Performance

As well as considering correlation within the design matrix as a factor which impacts lasso performance, we can also consider the high dimensionality of the data, the size of parameters in the true relationship (4) and the noise strength of ϵ.

We conducted our tests on a high dimensional $p = 400$ dataset with only $n = 200$ samples. In order to improve our understanding of the performance of lasso regression on high dimensional datasets, we could compare with other regression methods such as OLS. We claim that the automatic feature selection as a result of the weighted l^1 norm in lasso will keep performance stable in high dimensions [1], as is suggested by our current results.

Another consideration for the performance of lasso regression is the size of the β parameters. If the size of entries in the true regression parameter β in (4) are small, lasso is more likely to dismiss the predictors as random noise and fail

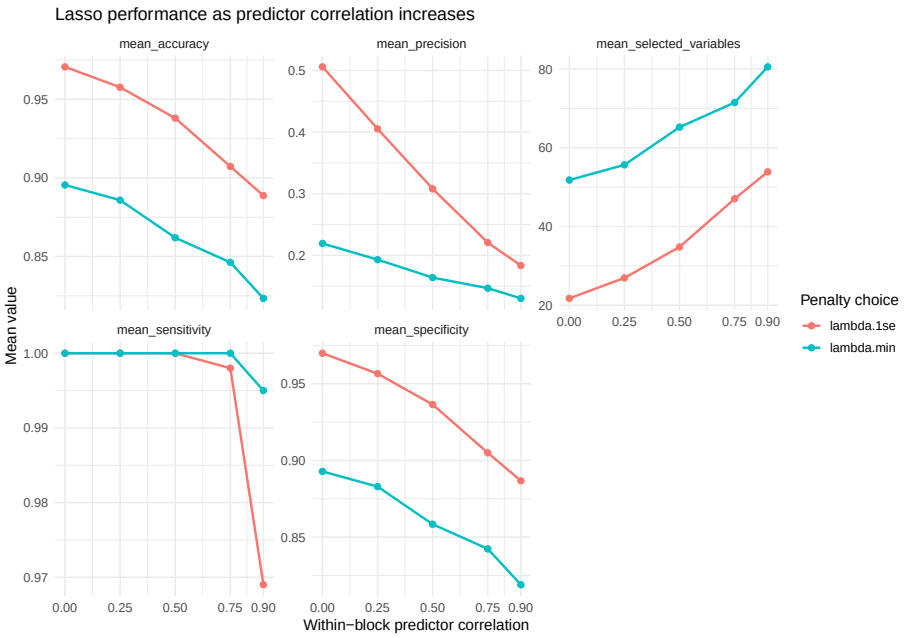


Figure 10: Average performance metrics over 100 runs of lasso regression for varying choices of within block correlation in the design matrix. Values are plotted for both the most parsimounious choice of λ (lambda.1se) and the λ which minimised error on the training data (lambda.min).

to identify the true associated variables [3] while for larger entries in the true β, lasso is more easily able to separate truly associated variables from correlated features. We could repeat our tests across a range of true regression parameters β in order to examine the impact of this relationship on lasso performance for different NLMs with a range of continuous response vectors.

Finally, we note that in the constructed NLM (4) the noise ϵ is assumed to be normally distributed with unit variance. For smaller variance, the lasso may be more accurate in identifying associated variables, while for larger values of ϵ, lasso may struggle to distinguish true relationships in the data from random noise. By repeating our experiment across a range of variances for ϵ, or across a range of more general linear models we can establish the effect of noisy data on lasso performance. We claim that normally distibuted noise with smaller variance will allow lasso to better select features which are truly associated with the response, leading to sparser regression parameters and better prediction performance.

References

[1] P. Bühlmann and S. van de Geer. *Statistics for High-Dimensional Data: Methods, Theory and Applications*. Springer Series in Statistics. Springer, Berlin, Heidelberg, 2011.

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[3] MCP Analytics Team. Lasso regression: Practical guide for data-driven decisions. `https://mcpanalytics.ai/articles/lasso-regression-practical-guide-for-data-driven-decisions`, Mar. 2026. Updated October 29, 2026; Accessed November 6, 2026.

[4] F. Santosa and W. W. Symes. Linear inversion of band-limited reflection seismograms. *SIAM Journal on Scientific and Statistical Computing*, 7(4):1307–1330, 1986.